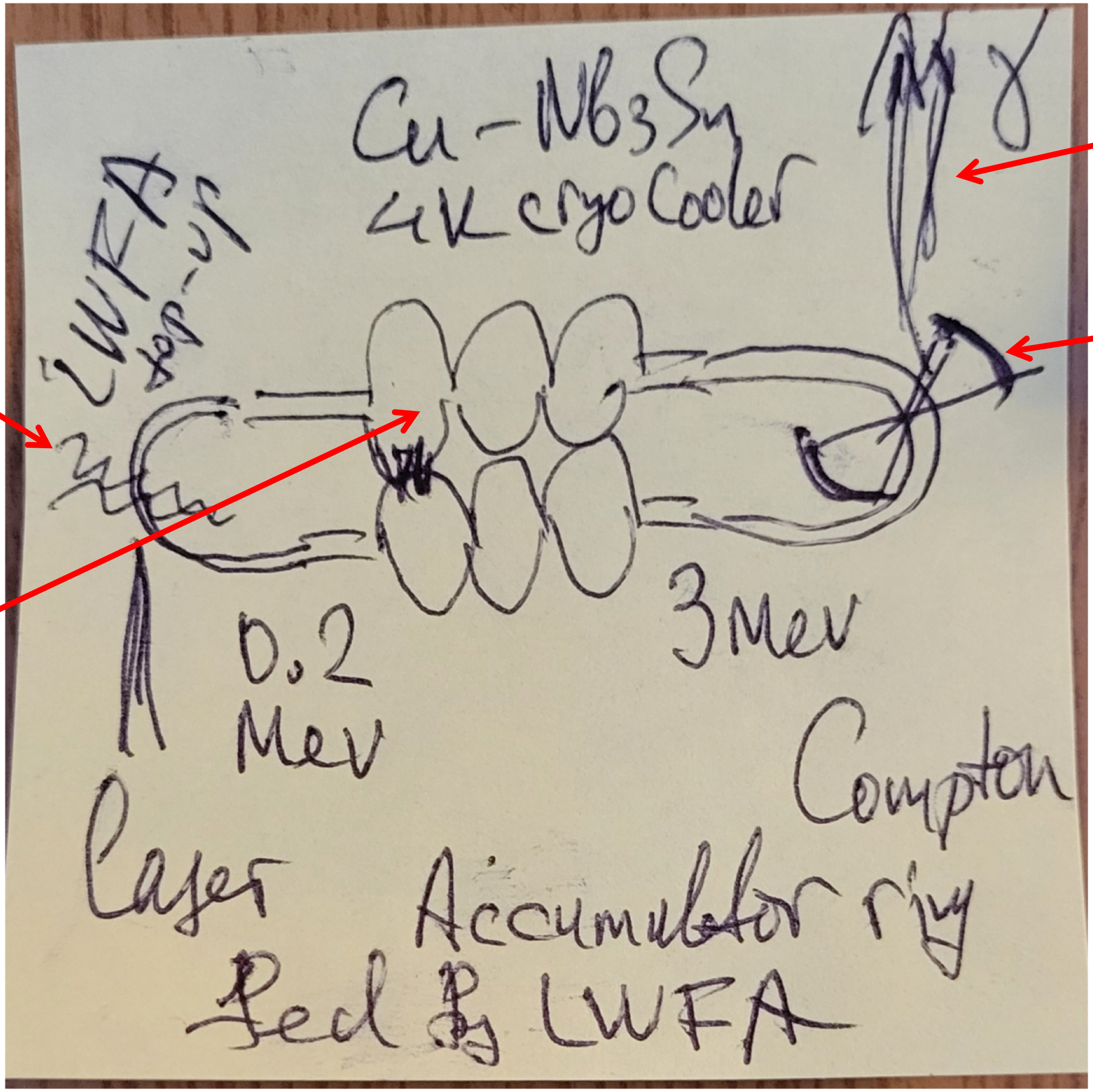
LWFA-fed Compton Source Based on Dual-Energy Accumulator Ring with Cryocooled Nb3Sn Twin-Axis Cavities

Laser plasma acceleration for top-up injection. Pulsed (fs) laser, pulsed gas jet target. Injection at about 0.2 MeV

Twin-axis SC cavities, copper covered by Nb3Sn, working at 4K, cooled by mobile cryocooler. Bottom cells accelerate beam, top cells decelerate beam. Energy gain about 1MeV per cell. Three cells give 3 MeV.



Compton photons with wavelength ~3 nm (inside “water window”)

Laser cavity to store photons (of ~600nm wavelength) for Compton collisions

Hypothesis: top-up LWFA injection will allow accumulation of ~Ampere scale current in the ring, helping to boost the source brightness

Possible staging/building blocks:
- LWFA fs laser injector;
  - Pulsed gas jet target;
- Twin-cavities Nb 2K prototype;
  - Twin cavities Nb3Sn at 4K;
- Laser cavity for Compton;
- Diagnostics/Control/Integration;